

AIM:

To implement a RecyclerView in Android using Kotlin to display a list of items and handle item click events.

Procedure

1. Open Android Studio.
2. Create a new project (Empty Views Activity, Kotlin, API 21+).
3. Add RecyclerView dependency in build.gradle.
4. Design layout with RecyclerView in activity_main.xml.
5. Create a row layout (item_row.xml).
6. Create a data model class (Item.kt).
7. Create an Adapter class to bind data to RecyclerView.
8. Set LayoutManager and Adapter in MainActivity.kt.
9. Implement click handling using Toast.
10. Run the app and observe output.

Code**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="0dp"
        android:layout_height="0dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

item_row.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/txtItem"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="20dp"
    android:textSize="18sp"/>
```

Item.kt

```
data class Item(val name: String)
```

ItemAdapter.kt

```
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import android.widget.TextView
import android.widget.Toast
import androidx.recyclerview.widget.RecyclerView

class ItemAdapter(private val itemList: List<Item>) :
    RecyclerView.Adapter<ItemAdapter.ItemViewHolder>() {

    class ItemViewHolder(itemView: View) : RecyclerView.ViewHolder(itemView) {
        val textView: TextView = itemView.findViewById(R.id.txtItem)
    }

    override fun onCreateViewHolder(parent: ViewGroup, viewType: Int): ItemViewHolder {
        val view = LayoutInflater.from(parent.context)
            .inflate(R.layout.item_row, parent, false)
        return ItemViewHolder(view)
    }

    override fun onBindViewHolder(holder: ItemViewHolder, position: Int) {
        val item = itemList[position]
        holder.textView.text = item.name

        holder.itemView.setOnClickListener {
            Toast.makeText(
                holder.itemView.context,
                "Clicked: ${item.name}",
                Toast.LENGTH_SHORT
            ).show()
        }
    }

    override fun getItemCount(): Int = itemList.size
}
```

MainActivity.kt

```
import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
    }
}
```

```
val recyclerView = findViewById<RecyclerView>(R.id.recyclerView)

recyclerView.layoutManager = LinearLayoutManager(this)

val items = listOf(
    Item("Android"),
    Item("Kotlin"),
    Item("RecyclerView"),
    Item("Advanced Concepts"),
    Item("Click Handling")
)

recyclerView.adapter = ItemAdapter(items)
}
```

Result

The application successfully:

- Displays a list using RecyclerView
- Handles item click events
- Shows a Toast message on item click

